

AOI-Inspection — Standard Operating Procedure

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AOI-Inspection — Standard Operating Procedure

Audience: Quality Worker · Maintenance Tech · Line Manager
Applies to: Line-E, station 4, AOI-Inspection (Automated Optical Inspection)

1. Purpose

AOI-Inspection performs automated optical inspection of the soldered SMT joints leaving Reflow-Oven. Reject disposition follows [Quality_Reject_Disposition_Policy.pdf](#).

2. Normal operating envelope

- Ideal cycle time: **10 s**
- Max acceptable cycle time: **14 s**
- Expected reject rate: **$\leq 4\%$**
- Fault probability per cycle: **0.3 %**
- Input buffer capacity: **5 parts**

3. Idle reasons

idle_reason	Meaning here	First check
Starved	Reflow-Oven (station 3) is not feeding boards.	Verify Reflow-Oven state.
Blocked	Through-Hole-Insert (station 5) buffer full.	Verify Through-Hole-Insert state.

4. Faults & corrective actions

4.1 Camera-Fault

- **Symptoms in telemetry:** `fault_type = "Camera-Fault"`. Inspection cycle aborts or image acquisition fails.
- **Likely root cause:** lens fouled, focus drifted, camera failure.
- **Corrective action:**
 1. Clean lens; cycle camera power.

- 2. Replace camera if acquisition still fails.
- **Restart criteria:** master PCB passes recipe.

4.2 Light-Fail

- **Symptoms in telemetry:** fault_type = "Light-Fail". Image contrast drops.
- **Likely root cause:** ring-light LED bank failed.
- **Corrective action:**
 1. LOTO. Replace the light bar.
 2. Restore brightness setpoint.
- **Restart criteria:** image meets contrast threshold.

4.3 Calibration-Drift

- **Symptoms in telemetry:** fault_type = "Calibration-Drift". False-positive or false-negative rejects rise.
- **Likely root cause:** classifier drift; temperature-induced offset.
- **Corrective action:**
 1. Run the calibration set (known-good + known-defect PCBs).
 2. Tune thresholds; sign off recipe revision in the QMS.
- **Restart criteria:** 100 % accuracy on the calibration set.

5. Preventive maintenance schedule

- **Daily:** master PCB verification at shift start.
- **Weekly:** clean lenses and diffusers.
- **Monthly:** classifier retraining.
- **Quarterly:** lighting replacement.

6. References

- Quality_Reject_Disposition_Policy.pdf
- Shift_Handover_Checklist.pdf
- Line-E_03_Reflow-Oven_SOP.pdf
- Line-E_05_Through-Hole-Insert_SOP.pdf